



COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
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ACTIVITY NO. 2.1 SSH Key-Based Authentication and GIT Setup

1. PROCEDURE OUTPUT

Task 1:

```
keliste@Keliste-Workstation:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/keliste/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/keliste/.ssh/id_rsa
Your public key has been saved in /home/keliste/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:d5NC0o+ZM5f3l1LmItDyflIVqvM3qBuGbdsPIWNRLPE keliste@Keliste-Workstation
The key's randomart image is:
+----[RSA 4096]-----+
|      .+o      |
|      ...      |
|     .. E      |
|     oo .      |
|    S=+o+.     |
|   =Xo==..o    |
|  .*=0ooo+ .   |
| o=oB.B o.    |
|  +=o* =.     |
+-----[SHA256]-----+
keliste@Keliste-Workstation:~$
keliste@Keliste-Workstation:~$ ls -la ~/.ssh
total 24
drwx----- 2 keliste keliste 4096 Jul 23 10:17 .
drwxr-x--- 15 keliste keliste 4096 Jul 23 09:33 ..
-rw----- 1 keliste keliste 3389 Jul 23 10:17 id_rsa
-rw-r--r-- 1 keliste keliste 753 Jul 23 10:17 id_rsa.pub
-rw----- 1 keliste keliste 1956 Jul 23 09:34 known_hosts
-rw----- 1 keliste keliste 1120 Jul 23 09:34 known_hosts.old
keliste@Keliste-Workstation:~$
```

Task 2:

```
keliste@Keliste-Workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub keliste@192.168.56.103
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/keliste/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that a
re already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is t
o install the new keys
keliste@192.168.56.103's password:
```

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'keliste@192.168.56.103'"
and check to make sure that only the key(s) you wanted were added.

```
keliste@Keliste-Workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub keliste@192.168.56.104
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/keliste/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that a
re already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is t
o install the new keys
keliste@192.168.56.104's password:
```

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'keliste@192.168.56.104'"
and check to make sure that only the key(s) you wanted were added.

```
keliste@Keliste-Workstation:~$ ssh keliste@192.168.56.103
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 23 09:33:37 2026 from 192.168.56.102

```
keliste@Keliste-Server1:~$ exit
```

logout

Connection to 192.168.56.103 closed.

```
keliste@Keliste-Workstation:~$
```

```
keliste@Keliste-Workstation:~$ ssh keliste@192.168.56.104
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 23 09:34:42 2026 from 192.168.56.102
keliste@Keliste-Server2:~$ exit
logout
Connection to 192.168.56.104 closed.
keliste@Keliste-Workstation:~$
```

Task 3:

```
keliste@Keliste-Workstation:~$ which git
/usr/bin/git
keliste@Keliste-Workstation:~$ git --version
git version 2.34.1
keliste@Keliste-Workstation:~$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQADJNqG3CVX0sIe+zKzBLfDv8DQ01vWRWABkjDNEdstR
4+7v9p0k+uINC6YctOPcAhDes2ySLu84iAXi+PmEODkkdeUt3se66bZGMAjTtbeZag+UMtHaPoPCYJM+
k9dfJmhzaFLNXbsr3fr4yBDcso8vcTyIQ7SIEVMfBFZj/JvQJkWqPTv+KOWXjeBtOQ7qzRoamANYneXR
W6336iTw0Q+FKUeEdHRxJWgBkg8Nkxo0DYhEViGpTT7yygnbWRwVhricpHUockAfFnt8Bwnsjgp3uj64
qmgsnH7se/uzKC0IudZjcBgt+axLnZ0UfC11HnOEKUXjXShDLveHoS2DGa4ggFg8Wog927G3y60v//vk
/wG93L5/o/DArn+zs3zjLsUmmvZd0knBjRn+KVEfT2blHsCLIRYMJChyCs1lxPSodueH3j+w0wJapvUe
3imjzeAJDMib+8/VPQpqbL9+PHsF+wq+/YFg9sdttxB9KEwMrxt7v4YxHGcnIDnh9GAnbFvIeXXBa/y
wWnHasFMDaFbZOMu37ypUSuFFlgF2BFI4KQlVdMtnHpBg0y+l0MJ0XyA/dw00dq3L8rzkfyhh8W1HlkZ
7Yzj0UzEzMUEnIOUqJIzSKLn3cCg+PlFvZmEZ5nl0aFnDhRNC9uFuFKodBUKgTBfkyeWdStBaqaiZCsF
Ow== keliste@Keliste-Workstation
```



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Pin

Watch 0

Fork 0

Star 0

main 1 Branch 0 Tags

Go to file

T

+

Code

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

README.md

Initial commit

21 minutes ago

README

CPE212_KELISTE

```
git@github.com: AiconKeliste/CPE212_KELISTE.git: No such file or directory
keliste@Keliste-Workstation:~$ git clone git@github.com:AiconKeliste/CPE212_KELISTE.git
Cloning into 'CPE212_KELISTE'...
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
keliste@Keliste-Workstation:~$
```

```
keliste@Keliste-Workstation:~$ cd CPE212_KELISTE
keliste@Keliste-Workstation:~/CPE212_KELISTE$ LS
LS: command not found
keliste@Keliste-Workstation:~/CPE212_KELISTE$ ls
README.md
keliste@Keliste-Workstation:~/CPE212_KELISTE$
```

```
keliste@Keliste-Workstation:~/CPE212_KELISTE$ git config --global user.name AiconKeliste
keliste@Keliste-Workstation:~/CPE212_KELISTE$ git config --global user.email qahkeliste@tip.edu.ph
keliste@Keliste-Workstation:~/CPE212_KELISTE$ cat ~/.gitconfig
[user]
    name = AiconKeliste
    email = qahkeliste@tip.edu.ph
keliste@Keliste-Workstation:~/CPE212_KELISTE$
```

```

keliste@Keliste-Workstation: ~/CPE212_KELISTE
GNU nano 6.2 README.md *
CPE212_KELISTE
This for hands-on-activity 2 for CPE212

[ Read 2 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line

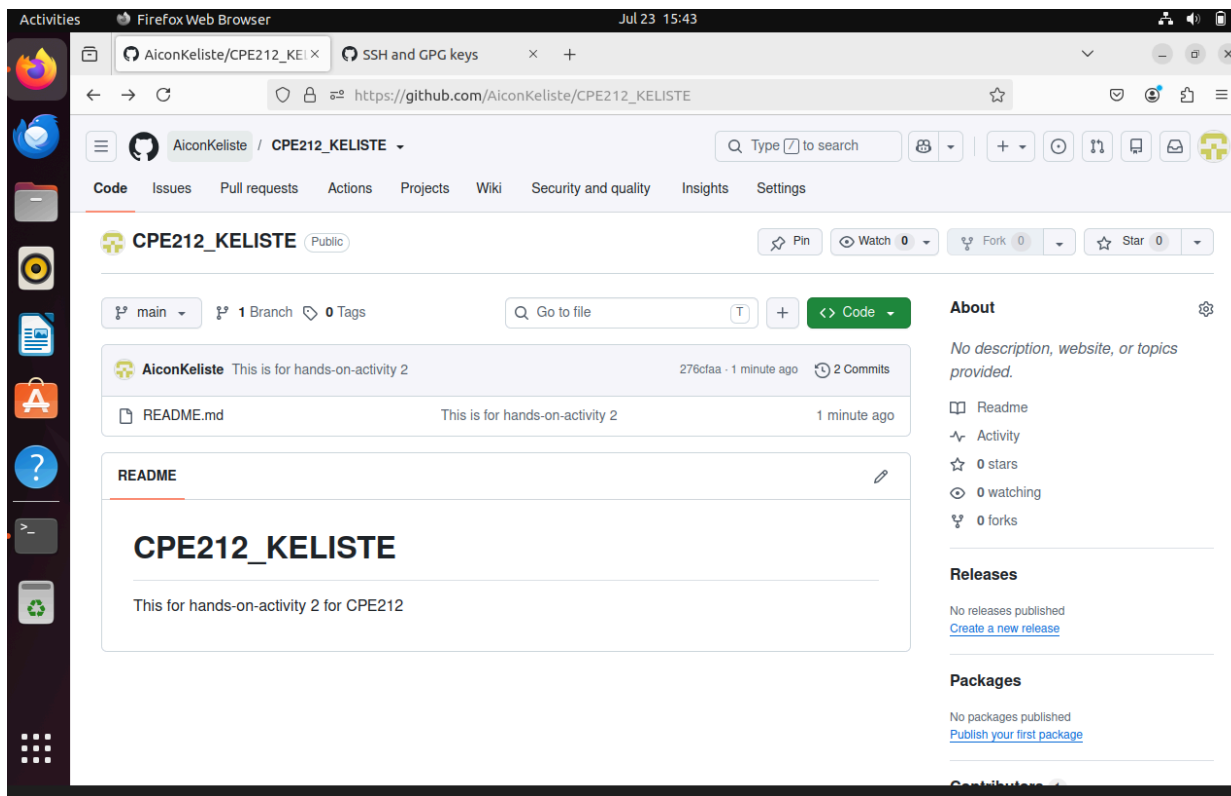
keliste@Keliste-Workstation:~/CPE212_KELISTE$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
keliste@Keliste-Workstation:~/CPE212_KELISTE$

keliste@Keliste-Workstation:~/CPE212_KELISTE$ git add README.md
keliste@Keliste-Workstation:~/CPE212_KELISTE$ git commit -m This is for hands-on-activity 2
error: pathspec 'is' did not match any file(s) known to git
error: pathspec 'for' did not match any file(s) known to git
error: pathspec 'hands-on-activity' did not match any file(s) known to git
error: pathspec '2' did not match any file(s) known to git
keliste@Keliste-Workstation:~/CPE212_KELISTE$ git commit -m "This is for hands-on-activity 2"
[main 276cfaa] This is for hands-on-activity 2
 1 file changed, 2 insertions(+), 1 deletion(-)
keliste@Keliste-Workstation:~/CPE212_KELISTE$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 315 bytes | 315.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:AiconKeliste/CPE212_KELISTE.git
   b6362a1..276cfaa  main -> main
keliste@Keliste-Workstation:~/CPE212_KELISTE$

```



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. How will you describe the ssh-program? What does it do?
 - SSH is a network protocol that allows the user to connect and control a remote computer over an unsecured network. Unlike other protocols, ssh encrypts all communication such as passwords and datas.
2. How do you know that you already installed the public key to the remote servers?
 - You can check the remote server by running `cat ~/.ssh/authorized_keys` which will display the public key.
3. What sort of things have we so far done to the remote servers using ansible commands?
 - Ansible ad-hoc commands to test connectivity such as `ansible all -m ping` that can be used to ping another server or system.
4. How important is the inventory file?
 - Inventory file is important because it tells ansible exactly which remote servers to manage.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Chatgpt

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Idea generation

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.



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Extent of Use: Used for idea generations and also for grammar checking

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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